



FENICSx IN METAL ADDITIVE MANUFACTURING

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The talk covers the application of FEniCSx to the simulation of a metal 3D printing process, namely wire arc additive manufacturing (WAAM). The focus will be on the definition of moving heat sources and element activation strategies. It is demonstrated how simulation can help analyze temperatures, distortions, and microstructural evolution during metal printing, thus improving the quality and efficiency of metal additive manufacturing.